

KANDALAKSHA, Russian Federation

WMO: 222170

Lat: 67.1692N

Lon: 32.3542E

Elev: 83

StdP: 14.65

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.8	-15.4	-26.2	1.3	-21.7	-19.7	1.9	-15.4	17.3	20.1	14.9	17.8	3.5	0	0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.1	74.3	61.9	70.9	60.2	67.8	58.8	64.2	71.3	62.0	68.4	60.1	65.9	6.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	81.0	66.9	59.2	75.4	65.1	57.2	70.0	63.4	29.4	71.1	27.9	68.5	26.5	66.2	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.1	12.2	10.9	-26.1	81.3	6.7	4.0	-31.0	84.2	-34.9	86.5	-38.7	88.7	-43.5	91.6		
			-26.3	66.8	6.7	2.8	-31.1	68.8	-35.1	70.4	-38.8	72.0	-43.7	74.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	33.4	10.5	11.3	20.3	30.8	41.2	52.1	58.6	54.5	45.6	34.8	23.3	16.5	(6)
(7)		DBStd	18.74	13.42	13.32	9.73	7.50	6.84	6.55	5.93	5.90	5.96	7.28	9.96	12.21	(7)
(8)		HDD50	6632	1223	1084	921	575	285	51	5	20	158	471	801	1038	(8)
(9)		HDD65	11541	1688	1504	1386	1025	736	390	210	330	583	935	1251	1503	(9)
(10)		CDD50	587	0	0	0	0	14	114	273	160	26	0	0	0	(10)
(11)		CDD65	21	0	0	0	0	0	3	13	5	0	0	0	0	(11)
(12)		CDH74	127	0	0	0	0	1	23	65	38	0	0	0	0	(12)
(13)		CDH80	15	0	0	0	0	0	5	5	5	0	0	0	0	(13)
(14)	Wind	WSAvg	5.2	5.0	5.1	5.4	5.3	5.8	5.3	4.9	5.0	4.9	5.0	5.1	(14)	
(15)	Precipitation	PrecAvg	20.0	1.4	1.0	1.1	0.9	1.5	2.0	2.6	2.3	2.2	2.0	1.7	1.3	(15)
(16)		PrecMax	27.2	3.0	2.2	2.4	2.1	3.5	4.3	5.7	4.8	4.3	5.0	4.1	2.5	(16)
(17)		PrecMin	14.3	0.2	0.3	0.2	0.0	0.2	0.4	0.0	0.7	0.4	0.3	0.3	0.3	(17)
(18)		PrecStd	3.1	0.7	0.5	0.6	0.5	0.8	1.0	1.3	0.8	1.1	1.0	0.7	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.6	38.7	43.8	53.0	68.7	76.9	80.2	79.2	64.9	52.4	44.3	38.4	(19)
(20)		0.4%	MCWB	33.6	36.0	37.2	43.3	54.1	61.4	64.8	65.4	56.7	49.0	42.4	35.3	(20)
(21)		2%	DB	33.2	34.3	38.7	47.3	61.6	70.5	75.2	72.3	60.6	49.6	40.5	35.9	(21)
(22)		2%	MCWB	32.2	32.6	34.1	40.0	49.9	58.1	62.3	61.4	54.0	46.9	39.1	34.3	(22)
(23)		5%	DB	30.8	32.1	35.6	43.8	57.0	66.6	72.2	67.9	57.7	47.0	37.2	33.8	(23)
(24)		5%	MCWB	29.8	30.7	32.5	37.4	47.5	55.5	61.3	60.1	52.8	45.0	36.2	32.7	(24)
(25)		10%	DB	28.5	29.5	33.4	41.1	53.1	63.3	69.5	64.5	55.3	44.7	35.0	31.8	(25)
(26)		10%	MCWB	27.7	28.3	30.9	35.8	45.7	53.8	60.0	58.3	51.7	43.0	34.1	30.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.2	36.2	37.5	44.2	54.7	63.8	68.1	67.5	57.7	49.8	42.5	36.4	(27)
(28)		0.4%	MCDWB	34.9	38.4	42.6	51.4	64.7	73.5	74.9	76.7	62.7	51.5	43.7	38.0	(28)
(29)		2%	WB	32.4	33.0	35.0	40.5	51.0	60.0	65.1	63.3	55.4	47.2	39.2	34.5	(29)
(30)		2%	MCDWB	33.0	34.1	37.8	46.1	59.3	67.8	72.0	69.6	59.1	48.9	40.3	35.6	(30)
(31)		5%	WB	29.9	30.9	33.0	38.3	48.5	57.2	63.0	60.8	53.7	45.2	36.3	32.8	(31)
(32)		5%	MCDWB	30.8	32.0	35.1	42.7	55.1	64.3	69.6	66.4	57.0	46.8	36.9	33.8	(32)
(33)		10%	WB	27.6	28.4	31.3	36.6	46.2	55.2	61.2	58.9	52.0	43.2	34.2	31.0	(33)
(34)		10%	MCDWB	28.5	29.5	33.4	40.1	52.5	61.8	67.2	63.7	54.6	44.6	34.9	31.9	(34)
(35)	Mean Daily Temperature Range		MDBR	12.9	14.0	15.7	14.7	13.3	14.0	14.1	13.7	12.3	8.7	9.0	11.4	(35)
(36)		5% DB	MCDBR	10.5	10.8	14.6	17.8	20.6	19.9	19.7	19.0	16.1	9.8	7.3	8.7	(36)
(37)			MCWBR	10.1	9.8	11.4	11.3	11.1	9.8	9.2	9.8	10.7	7.7	6.7	7.8	(37)
(38)			MCDBR	10.1	10.7	12.6	15.7	18.5	16.4	16.4	16.1	13.5	8.7	7.0	8.2	(38)
(39)		5% WB	MCWBR	9.8	9.9	10.2	10.5	10.9	9.2	8.8	9.3	9.8	7.3	6.6	7.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.209	0.269	0.304	0.338	0.356	0.355	0.374	0.353	0.323	0.286	0.358	N/A	(40)	
(41)		taud	1.923	2.063	2.117	2.199	2.372	2.445	2.428	2.460	2.517	2.300	2.649	N/A	(41)	
(42)		Ebn at Noon	44	170	231	257	266	270	259	253	235	170	54	0	(42)	
(43)		Edh at Noon	4	17	28	34	32	30	30	26	19	13	5	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	18	166	612	1130	1453	1624	1479	1037	561	185	34	0	(44)	
(45)		RadStd	1	27	60	86	117	141	182	96	53	27	4	0	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (14 neighbors)	+1.08	N/A	N/A	N/A	N/A	N/A	-239	-360	N/A	N/A
		+0.99	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page